23 Carlo puts tins into small boxes and into large boxes.

He puts 6 tins into each small box.
He puts 20 tins into each large box.

Carlo puts a total of 3000 tins into the boxes so that

number of tins in small boxes : number of tins in large boxes = 2 : 3

Carlo says that less than 30% of the boxes filled with tins are large boxes.

Is Carlo correct?
You must show all your working.

Five ratio parts = 3000 tins, so one part = 600.

Small-box tins: 2 × 600 = 1200; 1200 ÷ 6 = 200 boxes.

Large-box tins: 3 × 600 = 1800; 1800 ÷ 20 = 90 boxes.

Large-box share = 90 ÷ (200 + 90) × 100 = 31.0%.

No: 31.0% is more than 30%.

5 marks total
600 per ratio part
Split 1200 : 1800
200 small, 90 large
90/290 × 100
No; about 31%

(Total for Question 23 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA